

Toward a Tractability Frontier for Exact Relevance Certification

Tristan Simas
McGill University
`tristan.simas@mail.mcgill.ca`

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Abstract

Exact relevance certification asks which coordinates are necessary to determine the optimal action in a coordinate-structured decision problem. We first give a positive structural consolidation: the currently formalized tractable landscape splits into core mechanisms, regime lifts, and degenerate collapses, yielding a finite primitive basis for the present artifact. We then prove a maximal realizability barrier: arbitrary labeling kernels are optimizer-realizable, so quotient shape alone cannot characterize a tractability frontier.

The main new result is a meta-impossibility theorem for a principled admissibility class. Admissible predicates are required to be polynomial-time checkable, structural, and invariant under closure laws already forced by exact-certification semantics (relabelings, positive affine transformations, and duplication/irrelevance operations). Within this class, we build explicit same-orbit disagreement witnesses for four obstruction families—dominant-pair concentration, margin masking, ghost-action concentration, and additive/statewise offset concentration. Each witness uses an action-independent, pair-targeted affine term to move within a closure orbit while changing obstruction status. Consequently, no admissible normalization predicate can provide an exact tractability characterization over these families. This yields a formal admissible-collapse theorem and places the frontier program in a Rice/Arrow-style meta-impossibility tradition: once structural invariances are accepted, the sought classifier cannot exist in the naive finite form.

1 Introduction

Exact relevance certification asks: given a coordinate-structured decision problem, which coordinates are necessary to determine the optimal action set? This question appears whenever one seeks principled feature deletion or abstraction while preserving optimization behavior. Under succinct encodings, exact certification is hard in the ambient regimes (coNP-complete in static settings and PSPACE-complete in sequential settings), so any tractability statement must be genuinely structural.

This paper should be read as the meta-level companion to the optimizer-quotient and certification-trilemma line of work [18]. That line isolates the semantic object that exact certification depends on and maps the surrounding complexity landscape across regimes. The present paper asks a different question: once that semantic core is fixed, can one still hope for a finite structural tractability classifier at the representation level? The answer established here is negative for the admissibility class developed below.

The current formal landscape is already rich enough to separate two distinct tasks. The first is “positive consolidation”: identify and organize known polynomial-time islands. The second is

“meta-frontier control”: determine whether a finite structural classifier can exist at all once one respects the invariances forced by exact-certification semantics. This paper addresses both tasks.

On the positive side, we prove that the currently formalized tractable families split into three roles: primitive core mechanisms, regime lifts, and degenerate collapses. This yields a finite primitive basis for the present artifact. On its own, however, this does not establish a frontier theorem: finite inventory is not finite characterization.

The natural next move is to classify by optimizer-induced quotient structure. We show this is impossible in the unconstrained sense: optimizer realizability is maximal, and arbitrary labeling kernels are realizable. Thus quotient shape is too expressive to carry the boundary by itself.

This realizability barrier motivates the organizing principle of the paper: tractability predicates must be evaluated on representation-level structural classes, but modulo theorem-forced equivalences of presentation. In particular, closure under relabelings, positive affine utility reparameterizations, duplication, and irrelevant-coordinate extension is not a modeling preference; it is inherited from the fact that exact certification depends only on the induced decision quotient relation.

At the same time, unrestricted predicates trivialize any impossibility theorem via oracle encoding. We therefore isolate a principled admissibility layer: polynomial-time checkability, structural extractability, closure-law invariance, and bounded-pattern definability. This class excludes direct semantic encoding while still covering the formal results proved here.

Within this admissibility class, we prove a four-family no-go template. For each obstruction family we construct two slices with different obstruction status in the same closure orbit. The key mechanism is an action-independent, pair-targeted additive affine term (the statewise “ α -component” of positive affine transport), which changes the obstruction predicate while preserving closure equivalence. Closure-law invariance then transports any candidate predicate across the orbit and forces contradiction.

The dominant-pair family is the canonical instance, and the same mechanism now applies to margin masking, ghost-action concentration, and additive/statewise offset concentration. The resulting theorem states that no admissible normalization predicate can serve as a tractability characterization in the required sense.

Conceptually, the result belongs to the same meta-impossibility lineage as Rice-style and Arrow-style theorems: once one accepts a natural invariance package and a nontrivial expressivity domain, the desired classifier cannot exist in the naive finite form. What remains open is not whether arbitrary finite classifiers fail, but what stronger structural principle (if any) yields a correct frontier theorem.

Roadmap. Section 2 gives the finite-basis consolidation for the current artifact; Section 3 separates lifts from degeneracies; Sections 4–4c develop the realizability barrier, admissibility motivation, and obstruction-family no-go program; Section 5 positions the result in related meta-impossibility and tractability traditions; Section 6 concludes.

2 Finite Basis for the Current Artifact

We distinguish between *formalized tractable families* and *primitive tractability mechanisms*. A formalized family is any tractable condition with its own theorem statement or formal module in the current development. A primitive mechanism is a smaller conceptual source of tractability that may explain several families at once.

For the present artifact, the primitive list is explicit:

1. the six core structural subcases,

2. constant-optimizer collapse, and
3. finite explicit enumeration.

Thus the current primitive basis has size eight.

Theorem 2.1¹ (Explicit Partition of the Current Formalized Landscape). *The fifteen currently formalized tractable families admit an explicit disjoint partition into three blocks:*

1. *six core structural families,*
2. *four lifted families that reduce to existing core subcases, and*
3. *five degenerate families that become tractable by optimizer collapse or finite explicit enumeration.*

Moreover, every currently formalized family is classified into exactly one of these three blocks by an explicit formal classification map.

Proposition 2.2² (Explicit Inventory). *The fifteen currently formalized families are exactly the following:*

1. *core: bounded actions, separable utility, low tensor rank, tree structure, bounded treewidth, coordinate symmetry;*
2. *lifts: product distribution, bounded support, bounded horizon, full observability;*
3. *degenerate: single action, bounded state space, strict global dominance, constant optimal set, multiplicative-separable constant-sign.*

Corollary 2.3³ (Finite Basis for the Current Mechanized Landscape). *Projecting the partition of Theorem 2.1 to primitive mechanisms yields an explicit finite list of eight primitive tractability mechanisms. The list consists of the six core structural subcases*

bounded actions, separable utility, low tensor rank, tree structure, bounded treewidth, coordinate symmetry, together with two degenerate mechanisms:

constant-optimizer collapse and finite explicit enumeration.

Proposition 2.4⁴ (Explicit Primitive Basis). *The current primitive basis consists exactly of bounded actions, separable utility, low tensor rank, tree structure, bounded treewidth, coordinate symmetry, constant-optimizer collapse, and finite explicit enumeration.*

Proposition 2.5⁵ (Partition Counts). *The three blocks from Theorem 2.1 have sizes*

$$6 \text{ (core),} \quad 4 \text{ (lifted),} \quad 5 \text{ (degenerate).}$$

The current formalized landscape therefore contains fifteen tractable families accounted for by eight primitive mechanisms.

Proposition 2.6⁶ (All Independent Core Mechanisms Are Nondegenerate). *The current formal development already contains witness families showing that five core mechanisms are individually indispensable in the sense of basis coverage:*

¹ Lean: [FR1](#), [FR2](#) ² : [FR25](#), [FR26–28](#) ³ : [FR3](#) ⁴ : [FR29](#) ⁵ : [FR2](#) ⁶ : [FR101](#), [FR105](#), [FR106](#), [FR108](#), [FR112](#), [FR113](#)

1. a low-rank-only witness family,
2. a separable-only witness family,
3. a bounded-actions-only witness family,
4. a bounded-treewidth-only witness family, and
5. a symmetry-only witness family.

Each of these families remains tractable by its designated mechanism while failing the currently formalized versions of the other independent core mechanisms.

Proposition 2.6 now settles the independent part of the basis question on the stable formal surface. Every core mechanism that is not merely contained in another is formally known to be genuinely needed.

More concretely, the current stable formal surface already contains five distinct kinds of “only-by” witness:

1. low-rank only,
2. separable only,
3. bounded-actions only,
4. bounded-treewidth only, and
5. symmetry only.

Proposition 2.7⁷ (Weak Tree Ordering Does Not Imply Width One). *There exists a dependency presentation satisfying the older monotone tree-order predicate **TreeStructured** whose dependency graph has real treewidth greater than one. Width-one containment requires the stricter parent-tree condition from Proposition 2.9, not merely monotone ordering.*

Proposition 2.8⁸ (Cyclic Dependencies Recover Hardness). *The reduction family for exact relevance certification yields **coNP**-hard instances with cyclic dependency structure. Thus the tree-structured regime is not only sufficient for tractability; leaving it also reaches the hard side.*

Proposition 2.9⁹ (Parent-Tree Structure Gives Width-One Decompositions). *Suppose a dependency presentation is tree-structured in the strict parent sense: every dependency points to a smaller coordinate and every coordinate has at most one parent. Then the induced dependency graph admits a genuine tree decomposition of width at most one.*

Proposition 2.9 is the mathematically honest containment theorem behind the intended “tree structure is the treewidth-one special case” slogan. The formal development distinguishes this parent-tree notion from the weaker monotone-order predicate currently named **TreeStructured**.

Proposition 2.10¹⁰ (Current Positive Interfaces Factor Through Role Classification). *The current positive theorem surface factors through the role classifier of Theorem 2.1. Concretely, the family-indexed interfaces for*

1. subcase-facing tractability,
2. degenerate-case witnesses, and
3. complexity-class summaries

⁷ : FR114 ⁸ : FR115 ⁹ : FR100, FR110, FR111 ¹⁰ : FR4

all factor through the map

$$\text{family} \longmapsto \text{role} \in \{\text{core}, \text{lifted}, \text{degenerate}\}.$$

Proposition 2.10 is the formal reason the current landscape feels smaller than its file count. The exported positive API is already controlled by the role partition rather than by fifteen unrelated theorem schemas.

3 Lifts and Degeneracies

The extra named families beyond the six core subcases split naturally into two groups.

Regime Lifts

Four stochastic/sequential positive results do not introduce new primitive mechanisms. They reduce directly to existing core subcases:

1. product distributions reduce to separable utility;
2. bounded support reduces to bounded actions;
3. bounded horizons reduce to bounded treewidth;
4. full observability reduces to tree structure.

These are mathematically useful because they transfer the static tractable theory into richer regimes. But they do not enlarge the primitive basis. They are best understood as *lifts*, not new frontier points.

Proposition 3.1¹¹ (Lifted Cases Introduce No New Primitive). *Each currently mechanized stochastic/sequential tractable case is classified into one of the six core structural subcases. In particular, the current stochastic/sequential positive results do not enlarge the primitive basis from Corollary 2.3.*

Degenerate Cases

The remaining currently mechanized positive cases are degenerate in a precise sense.

Single-action systems, strict global dominance, constant optimal-set systems, and multiplicative-separable constant-sign models all collapse the optimizer so strongly that the relevant-coordinate question disappears: the optimal-action set is constant across states, so the empty coordinate set is sufficient. These are not new structural tractability mechanisms for the relevance problem; they are constant-optimizer collapses.

Bounded state space behaves differently. It does not force a constant optimizer. Instead, it makes exact certification feasible by brute-force enumeration over a finite universe. This is polynomial only because the instance family already carries a hard external bound on its search space.

Proposition 3.2¹² (Degenerate Positive Cases). *Every currently mechanized non-core tractable case that is not a regime lift belongs to one of two degenerate buckets:*

1. *constant-optimizer collapse, or*
2. *finite explicit enumeration.*

Taken together with Proposition 3.1, Proposition 3.2 sharpens Theorem 2.1: every currently formalized positive result is exactly one of three things, a primitive core case, a lift, or a degenerate collapse.

¹¹ : FR5 ¹² : FR1, FR6

4 Toward the Frontier Theorem

Sections 2 and 3 suggest a natural hope. Perhaps the present tractable families already point toward a finite optimizer-compatible criterion. Perhaps every future positive result will still factor into primitive mechanisms, lifts, and degeneracies. In the strongest form, perhaps exact relevance certification admits a representation-sensitive dichotomy theorem analogous in spirit to the classical dichotomy programs for satisfiability and constraint satisfaction.

The present section explains why that hope cannot be maintained in this naive form. The realizability barrier shows that quotient shape is too expressive, while the admissibility-layer collapse theorem later in the section shows that even structurally restricted finite classifiers fail on a natural theorem-forced surface. What survives is not a finite classifier, but a sharper statement of what any future frontier theorem would have to avoid.

Three structural obstacles therefore sit between the current positive theory and any genuine metatheorem.

First: Realizability

Theorem 5.1 shows that naive realizability is not the bottleneck. In the unconstrained setting, arbitrary labeling kernels already arise as optimizer quotients. The frontier theorem therefore cannot come from quotient realizability alone. It has to characterize optimizer-compatible structure at the level of representation-level structural classes, not merely at the level of abstract set partitions. Section 6 isolates the first necessary conditions any such statement must satisfy.

Second: Primitive Versus Derived Tractability

A genuine frontier theorem must distinguish primitive mechanisms from lifts and degenerate cases, and show that this distinction is not an accident of the current examples.

Third: Algebraic Description

The natural long-term analogy is the CSP dichotomy program, where tractability is controlled by an algebraic boundary condition rather than by an ad hoc list of examples. For exact relevance certification, the unresolved question is whether optimizer-induced quotients admit a comparably finite algebraic description or whether the tractable side remains genuinely open-ended.

A Stabilized Binary Pairwise Subregime

One piece of the frontier program already admits a clean canonical theorem. On binary coordinate domains, define the mixed difference along coordinates i, j for action a by evaluating the utility on the four assignments $(0, 0), (1, 1), (0, 1), (1, 0)$ while holding all other coordinates fixed at 0. For pairwise utilities, this mixed difference is the canonical witness of genuine pair interaction.

Proposition 4.1¹³ (Binary Pairwise Symmetry Dichotomy). *Let the coordinate alphabet be binary. If a utility admits a pairwise decomposition and is invariant under coordinate permutations, then either:*

1. *it admits a unary-coordinate decomposition, so every pairwise term collapses into single-coordinate contributions, or*

¹³ : [FR118](#), [FR119](#), [FR120](#)

2. *every distinct coordinate pair carries a genuine pair interaction, witnessed by a nonzero mixed difference for some action.*

The point is structural rather than cosmetic. Within the binary pairwise-symmetric regime, there is no intermediate sparse interaction pattern hiding between unary collapse and the complete-pair case. Once a single nonzero pair witness exists, symmetry propagates it to every coordinate pair. This removes one plausible source of additional primitive mechanisms from the frontier search.

The optimizer-relevant version needs one extra layer. Raw pair interaction is still too coarse, because action-independent pairwise terms can be structurally dense without changing the optimizer. The correct invariant is the mixed difference of action gaps $U(a, \cdot) - U(b, \cdot)$.

Proposition 4.2¹⁴ (Decision-Relevant Binary Pairwise Dichotomy). *In the same binary pairwise-symmetric regime, either all action-dependent pair effects collapse into a unary-coordinate reduction after removing an action-independent base state term, or every distinct coordinate pair is decision-relevantly interacting for some action gap.*

This is the correct replacement for the naive hardness conjecture. The obstruction theorem FR124 shows that even complete genuine pair interaction and failure of coordinate symmetry do not force hardness by themselves: one can still have a constant optimizer. Any eventual frontier theorem must therefore quotient out optimizer-degenerate phenomena, not just declared or utility-level interaction density.

The obstruction is stronger than that first formulation suggests. Action-specific utility offsets leave the decision-relevant interaction graph unchanged FR125, yet they can turn a family with nonconstant optimizer behavior into one with a constant optimizer while preserving dense decision-relevant structure FR126, FR127. Any true hardness theorem in this regime must therefore normalize away action offsets explicitly, not merely require dense decision-relevant interaction.

This normalization can be stated cleanly: utilities are identified up to action-dependent, state-independent offsets, and the offset-normalized decision-relevant graph is well-defined on those equivalence classes FR128. But the resulting dichotomy attempt still fails. The offset-normalized obstruction theorem FR129 shows that even after passing to offset classes, one can retain complete decision-relevant interaction and unbounded treewidth while exact certification remains trivial.

The next refinement is to restrict the interaction graph to optimizer-supported actions. This removes both previously formalized collapse mechanisms: it removes the offset-collapse family and also a ghost-action family whose all-action decision-relevant graph is complete even though a single coordinate remains sufficient FR130, FR131, FR132, FR133. But even that support-filtered graph is not enough. The optimizer-supported obstruction theorem FR134, FR135, FR136 exhibits a margin-masking family with complete optimizer-supported decision-relevant interaction and unbounded treewidth, yet the optimizer still depends only on coordinate 0. The third collapse mechanism is therefore not ghost support or offset collapse, but large unary margins that mask dense supported pairwise interactions.

One might hope that a strict unary-to-pair margin bound rescues the dichotomy. Define *margin-bounded* pairwise utilities by requiring every unary term to be at most twice the largest binary mixed-difference magnitude FR137. This still fails. The dominant-pair family is margin-bounded because its unary terms vanish FR138, while its optimizer-supported decision-relevant graph remains complete FR139. Yet the obstruction theorem FR140 shows that exact certification is still controlled by the two-coordinate set $\{0, 1\}$. The fourth collapse mechanism is therefore pair-weight concentration rather than unary masking: one supported pair dominates the optimizer while the remaining dense interactions are too weak to matter.

¹⁴ : FR121, FR122, FR123, FR124

Admissible Closure-Orbit No-Go Program

The obstruction families above lead to a uniform negative statement. The key point is not that the families are difficult to recognize individually, but that they admit explicit disagreements inside a single closure orbit.

Proposition 4.3 (Orbit-Gap Template). *Let Q be a slice predicate. Suppose there exist slices U, V such that U and V lie in the same closure orbit, $Q(U)$ holds, and $Q(V)$ fails. Then no closure-law-invariant predicate P can satisfy*

$$P(W) \iff Q(W) \quad \text{for all slices } W.$$

Proof idea. Closure-law invariance gives $P(U) \iff P(V)$. Exact agreement with Q would then imply $Q(U) \iff Q(V)$, contradicting the assumed orbit gap. ■

The remaining work is therefore constructive: for each obstruction family, exhibit a same-orbit disagreement witness. In all four cases the witness is a positive-affine step whose α -component is an action-independent state term supported on a single coordinate pair. This is the technical hinge of the paper. Pure relabeling cannot change the relevant obstruction statistics, and a global scale factor preserves all pairwise magnitudes up to common rescaling. The statewise additive term is what allows one to move mass onto a chosen pair while remaining inside the theorem-forced closure laws. The fact that the same mechanism works for four structurally different obstruction families is the main evidence that the collapse theorem is detecting a genuine phenomenon rather than a collection of unrelated counterexamples.

Theorem 4.4 (Dominant-Pair Admissible No-Go). *No admissible normalization predicate can decide the dominant-pair obstruction family exactly.*

Proof sketch. The target predicate is unique dominant-pair status: one pair/action realizes the maximal interaction magnitude and that maximizer is unique. Start from a slice whose unique dominant pair is $\{0, 1\}$. Add an action-independent affine state term supported on a different pair, say $\{1, 2\}$. This produces a slice in the same closure orbit but with a different unique dominant pair. Proposition 4.3 then rules out every closure-law-invariant classifier, hence every admissible one. ■

Theorem 4.4 is the canonical instance. The remaining three families use the same orbit-gap scheme, but each isolates a different structural obstruction.

Theorem 4.5 (Margin-Masking Admissible No-Go). *No admissible normalization predicate can decide margin-boundedness exactly.*

Proof sketch. For margin masking, the target predicate is margin-boundedness itself. This family is structurally different from the dominant-pair family because it is governed by a threshold comparing unary mass against the largest pair interaction. The affine witness leaves unary terms unchanged but raises the largest pair interaction past the relevant threshold, so the translated slice becomes margin-bounded while the base slice is not. Proposition 4.3 therefore rules out every closure-law-invariant classifier, hence every admissible one. ■

Theorem 4.6 (Ghost-Action Admissible No-Go). *No admissible normalization predicate can decide the ghost-action concentration signature exactly.*

Proof sketch. For ghost actions, the mechanization uses a cast-free signature: there is an action whose unary contribution on the first coordinate is -1 on both binary values and whose anchor-pair interaction has unit mixed-difference magnitude. This family is local in appearance but unstable under the same pair-targeted affine move. A pair-supported affine term preserves the closure orbit while destroying the signature, so Proposition 4.3 applies. ■

Theorem 4.7 (Offset Admissible No-Go). *No admissible normalization predicate can decide the additive/statewise offset signature exactly.*

Proof sketch. For additive/statewise offset concentration, the cast-free target predicate requires two actions with distinct anchor-pair interaction magnitudes, namely 1 and 0. This family isolates the action-specific mismatch that survives after earlier offset normalizations. A pair-supported affine term changes the anchor-pair statistics while remaining in the same closure orbit. Again, Proposition 4.3 yields the contradiction. ■

Theorem 4.8 (Admissible Collapse Across the Four Obstruction Families). *Let P be a normalization predicate satisfying the admissibility axioms. Then P is not a tractability characterization.*

Proof idea. If P were an admissible tractability characterization, it would have to decide each of the four obstruction predicates above. Theorem 4.4, Theorem 4.5, Theorem 4.6, and Theorem 4.7 show that this is impossible. Hence no admissible normalization predicate yields the desired frontier theorem. ■

The mechanization exposes Theorem 4.8 with the formal signature

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admissibleCollapseLandscapeInfinity_full_paper :
forall (P : NormalizationPredicate),
AdmissibleNormalizationPredicate P ->
not (TractabilityCharacterization P).
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Internally, this theorem is obtained via a thin bridge from the working admissibility theorem over the four explicit obstruction predicates.

5 The Realizability Barrier

A natural hope for a finite tractability taxonomy would be this: perhaps optimizer-induced quotients realize only a narrow subclass of equivalence relations, so the finite frontier would emerge from realizability alone. In the unconstrained setting, that hope is false.

Theorem 5.1¹⁵ (Every Labeling Kernel Is Optimizer-Realizable). *Let $\phi : S \rightarrow T$ be any function. There exists a decision problem with action space T and state space S such that for every state $s \in S$,*

$$Opt(s) = \{\phi(s)\}.$$

Consequently, for all states $s, s' \in S$,

$$Opt(s) = Opt(s') \iff \phi(s) = \phi(s').$$

Equivalently, the decision quotient of the constructed problem is exactly the kernel partition of ϕ .

¹⁵ : FR7, FR8

Proof. Take the action set to be T itself and define

$$U(a, s) = \begin{cases} 1 & \text{if } a = \phi(s), \\ 0 & \text{otherwise.} \end{cases}$$

Then the unique maximizer at state s is the designated action $\phi(s)$, so the optimizer set is exactly $\{\phi(s)\}$. Two states therefore have the same optimizer set if and only if they receive the same label under ϕ . ■

Theorem 5.1 is the key negative fact for the frontier program. It shows that optimizer realizability by itself is extremely permissive. If arbitrary label kernels are already optimizer-induced, then no finite taxonomy can come merely from asking which abstract quotients are realizable.

Proposition 5.2¹⁶ (Every Equivalence Relation Is Optimizer-Realizable). *For every equivalence relation on the state space, there exists a decision problem whose decision quotient relation is exactly that equivalence relation. Moreover, the quotient object of the realizing problem is canonically equivalent to the corresponding setoid quotient.*

Proposition 5.2 is the maximal realizability statement available in the present framework. The labeling-kernel theorem is a special case. Thus the realizability barrier is not merely broad; at the level of abstract quotient relations, it is maximal.

Proposition 5.3¹⁷ (The Realized Quotient Is Exactly the Label Range). *For every labeling map $\phi : S \rightarrow T$, the decision quotient of the realizing problem is canonically equivalent to the range of ϕ . In finite settings, the quotient cardinality is therefore exactly $|\text{range}(\phi)|$.*

Proposition 5.3 strengthens Theorem 5.1. The realizing construction does not merely realize the induced equivalence relation abstractly; it realizes the quotient object itself with exactly the expected image size.

Corollary 5.4¹⁸ (Realizability Alone Cannot Be the Frontier). *Any future optimizer-compatible tractability metatheorem must use more than quotient realizability alone. Additional restrictions have to enter through representation, coordinate structure, utility structure, closure properties of the family under study, or comparable algebraic invariants.*

This shifts the burden of the metatheorem. The hard question is not which quotients can arise in principle; almost all kernels can. The hard question is which *natural structural families of decision problems* force tractability or preserve hardness once coordinate structure and utility representation are taken seriously.

6 Necessary Conditions on Any Frontier Statement

The realizability barrier leaves a more precise frontier question. If quotient shape alone is too expressive, what should replace it as the organizing unit of the metatheorem?

At minimum, the tractability frontier should not classify arbitrary sets of decision problems. It should range over representation-level structural classes and it should ignore benign changes of presentation.

The right eventual condition list will almost certainly be stronger than this minimal principle. For example, a mature frontier theorem may also require closure under adding explicitly irrelevant coordinates, adjoining dummy actions, or other representation-preserving refinements. But action/state relabel invariance is the irreducible starting point.

Proposition 6.1¹⁹ (Exact Certification Depends Only on the Decision Quotient Relation). *If two*

¹⁶ : FR45, FR46, FR47 ¹⁷ : FR39, FR40 ¹⁸ : FR7, FR8 ¹⁹ : FR30, FR31, FR32, FR33, FR34, FR35, FR37, FR51

decision problems on the same state space induce the same decision-equivalence relation, then they induce the same exact-certification structure. In particular, they have the same sufficient coordinate sets, the same minimal sufficient coordinate sets, and the same relevant and irrelevant coordinates. Moreover, their quotient objects are canonically equivalent; in finite settings, those quotient objects therefore have the same cardinality.

Proposition 6.1 is the canonical statement behind the frontier program. Exact certification is determined by the decision quotient relation itself. Equality of optimizer maps is only a sufficient route to this conclusion, because equal optimizer maps induce the same decision-equivalence relation. Corollary 6.2²⁰ (Sufficiency Is Relation Refinement). *A coordinate set I is sufficient if and only if the coordinate-agreement relation induced by I refines the decision quotient relation.*

Corollary 6.2 is the cleanest relational form of exact certification. It removes utility language entirely: sufficiency asks whether one state relation is contained in another.

Corollary 6.3²¹ (Relevance Is Failure of Erased Sufficiency). *A coordinate i is irrelevant if and only if deleting it leaves a sufficient coordinate set, namely $\text{univ} \setminus \{i\}$. Equivalently, i is relevant if and only if that erased set is not sufficient.*

Corollary 6.3 makes the role of individual coordinates canonical: relevance is exactly the obstruction to retaining sufficiency after coordinate erasure.

Corollary 6.4²² (Certification Statistics Factor Through the Quotient Relation). *Any statistic that is a function of the sufficient-set family and relevant-coordinate set is invariant under equality of the decision quotient relation. In particular, the number of sufficient coordinate sets, the number of minimal sufficient coordinate sets, and the number of relevant coordinates are quotient invariants.*

Corollary 6.4 records the next logical consequence of Proposition 6.1. Once exact certification is identified with quotient data, every certification statistic built from sufficient sets or relevant coordinates factors automatically through that quotient data as well.

Corollary 6.5²³ (Constant-Optimizer Collapse Forces Trivial Certification). *If the optimizer set is constant across all states, then the empty coordinate set is sufficient and every coordinate is irrelevant.*

Corollary 6.5 isolates the first genuinely degenerate mechanism in the tractable basis. The certification problem disappears because there is no state dependence left to certify.

Proposition 6.6²⁴ (Explicit Enumeration Is Parameter-Dependent, Not Structural). *For every fixed exponent k , there exists a Boolean-cube family with state space size $2^n > n^k$. Thus explicit-state enumeration can outrun any fixed monomial bound in the ambient dimension parameter.*

Proposition 6.6 is weaker than a full polynomial lower-bound metatheorem, but it already captures the structural point needed here: tractability by brute-force state enumeration depends on a separate size parameter and does not arise from an optimizer-compatible mechanism like symmetry, low rank, or tree structure.

Proposition 6.7²⁵ (Positive Affine Utility Reparameterizations Are Invisible). *Statewise positive affine transformations of utility leave exact certification unchanged. In particular, replacing*

$$U(a, s) \quad \text{by} \quad \alpha(s) + \beta(s)U(a, s)$$

with $\beta(s) > 0$ for every state s does not change sufficient coordinate sets or relevance judgments.

Proposition 6.7 is another direct consequence of Proposition 6.1. It records a standard utility-theoretic invariance in the exact-certification language: only the induced decision quotient matters, not the particular positive affine scale used to encode utility.

²⁰ : FR48 ²¹ : FR49, FR50 ²² : FR41, FR42, FR43, FR44 ²³ : FR52, FR53 ²⁴ : FR66, FR67 ²⁵ : FR13–16

Proposition 6.8²⁶ (Duplicate Actions and Duplicate States Preserve Certification). *Duplicating a single action without changing its utility profile preserves the decision quotient relation and hence preserves sufficiency and relevance. At the optimizer-set level, the only possible change is the addition of the duplicate action itself: original optimal actions remain optimal, and the duplicate is optimal exactly when the original action was. Duplicating a single state without changing its utility profile preserves the decision quotient up to the obvious quotient equivalence and also preserves sufficiency and relevance.*

Proposition 6.8 gives two more theorem-forced closure laws. These are not modeling conventions; they are consequences of the fact that exact certification factors through quotient data rather than through accidental multiplicity in the presentation.

Proposition 6.9²⁷ (Relabeling Invariance Is Forced by Exact Certification). *Action relabeling preserves optimizer equivalence and exact sufficiency. State relabeling does so as well once the coordinate structure is transported along the state bijection. Consequently, any structural tractability frontier for exact relevance certification must be closed under these relabelings.*

Proposition 6.9 is now best read as a corollary of Proposition 6.1. Bijective relabelings do not change the optimizer map up to the obvious identification, so they cannot change exact certification.

The second necessary ingredient is a genuine expressivity restriction. Call a class *kernel-universal* if, for every labeling map $\phi : S \rightarrow T$, it contains some decision problem whose optimizer quotient realizes exactly the kernel partition of ϕ . By Theorem 5.1, kernel universality is an extremely strong property.

The canonical finite invariant here is the size of the optimizer quotient, equivalently the number of distinct optimal-action sets. The quotient is identified with the range of the optimizer map, so quotient size is not auxiliary bookkeeping; it is intrinsic.

Proposition 6.10²⁸ (Quotient Size Is Unbounded Under Realizability). *For every $m \in \mathbb{N}$, there exists a finite decision problem whose optimizer quotient has size exactly m .*

Proof. Take $m = k + 1$ and realize the identity labeling on $\text{Fin}(k + 1)$. Proposition 5.3 identifies the quotient object with the range of that identity labeling, which has cardinality $k + 1$. Equivalently, Proposition 5.2 realizes the discrete equivalence relation on $\text{Fin}(k + 1)$. ■

The point is simple: a meaningful frontier statement cannot range over classes that are simultaneously relabel-invariant and kernel-universal, because quotient shape alone then becomes too expressive to isolate the boundary.

Proposition 6.11²⁹ (Invariance Alone Is Too Weak). *The universal class of all decision problems satisfies action relabeling invariance and state relabeling invariance, but it is kernel-universal. Consequently, relabeling invariance by itself is far too weak to isolate a tractability frontier.*

Proposition 6.11 separates two kinds of conditions that can otherwise be conflated.

1. **Benign invariance axioms** say the family should ignore arbitrary naming choices.
2. **Expressivity-limiting axioms** say the family should not realize arbitrary quotient structure wholesale.

Any genuine optimizer-compatible frontier theorem will need both. Without decision-quotient invariance and its immediate corollaries such as relabel invariance and positive affine invariance, the statement is not structural. Without an expressivity restriction excluding kernel universality

²⁶ : FR54, FR55, FR56, FR68, FR69, FR70, FR71, FR72, FR73 ²⁷ : FR17–20 ²⁸ : FR24, FR38 ²⁹ : FR21, FR22, FR23

or something comparably strong, the realizability theorem and Proposition 6.10 make the class too large for quotient shape alone to carry the boundary.

From Necessary Conditions to an Admissibility Class

The closure properties above are theorem-forced by exact-certification semantics, but they are not enough on their own to support a meaningful impossibility theorem. If one allows unrestricted predicates, a semantic predicate can simply encode the target complexity class and trivialize the statement.

Proposition 6.12 (Unrestricted Predicates Trivialize Exact Characterization). *Without admissibility restrictions, there exist normalization predicates whose membership condition directly encodes the intended polynomial-time side of the landscape, so unrestricted collapse-impossibility claims fail.*

Proposition 6.12 is not an artifact of the proof method. It is the expected behavior of unrestricted semantic predicates. Therefore any meaningful frontier theorem must be parameterized by an admissibility class that simultaneously (i) respects theorem-forced invariances and (ii) blocks direct semantic encoding.

Definition 6.13 (Admissible Normalization Predicate). A normalization predicate is *admissible* when it satisfies all of the following:

1. polynomial-time checkability at the slice level;
2. invariance under the closure laws forced by exact certification (relabelings, positive affine reparameterizations, duplication operations, and irrelevant-coordinate extension);
3. structural extractability of the associated dependency graph;
4. bounded-pattern definability.

Each clause in Definition 6.13 has an independent methodological role.

Polynomial-time checkability is the minimal algorithmic requirement. A tractability characterization that cannot itself be recognized efficiently has little explanatory value: it would only relocate the computational difficulty from exact certification to the membership test for the classifier.

Closure-law invariance is the decisive semantic requirement. It is not stipulated for convenience. It is forced by the Block 6 invariance theorems, because exact certification itself is unchanged by relabelings, positive affine utility reparameterizations, duplication operations, and irrelevant-coordinate extensions. The orbit-witness arguments of Section 4 use precisely this clause and no stronger semantic assumption.

Structural extractability distinguishes structural characterizations from purely semantic ones. Without such a condition, one can define predicates directly on optimizer behavior or on the solved certification instance itself, bypassing the stated aim of classifying tractability by representation-level structure.

Bounded-pattern definability is the locality guardrail. Its role is to exclude predicates whose membership test secretly performs an unbounded global computation under a structural name. In particular, it prevents one from reintroducing oracle power through an arbitrarily large finite schema.

Definition 6.13 is therefore not a list assembled solely to support the impossibility theorem. Closure-law invariance supplies the semantic content of the no-go theorem; the remaining clauses exclude vacuous or non-structural counterexamples and keep the classification problem within the intended algorithmic setting.

7 Related Work

Rough Sets, Feature Selection, and Exact Relevance

At the static level, the closest classical neighbor is rough-set reduct theory [13, 19]. That literature studies which attributes can be deleted while preserving decision distinctions. Exact static sufficiency can be viewed as a reduct condition for the induced decision table. The present paper is different in scope: it is not a new reduct algorithm, but a structural study of which tractable mechanisms have been formalized and whether they admit a finite characterization.

A separate neighboring literature studies feature selection, variable importance, and model explanation in machine learning and AI, including wrapper and filter methods [10], general surveys [9], and attribution methods based on Shapley-style decompositions [11]. These works are relevant by motivation rather than by technique. They typically optimize predictive performance, explanatory salience, or approximation quality under data-dependent criteria. Exact relevance certification instead asks for a semantic preservation property: which coordinates are necessary to preserve the optimizer relation exactly.

Backdoors, Tractable Islands, and Dichotomy Programs

The nearest complexity-theoretic analogue is backdoor tractability for constraint satisfaction [20, 3, 6]. There, a small structural set exposes membership in a tractable class even when the ambient problem is hard. The comparison is direct: exact relevance certification also asks which coordinates matter, and polynomial-time behavior emerges when the source of hardness is restricted by structure.

The broader analogy is the dichotomy tradition for satisfiability and constraint satisfaction, from Schaefer’s Boolean dichotomy theorem [16] to the finite-domain CSP dichotomy theorems of Bulatov and Zhuk [4, 23]. The expectation that a clean tractability boundary should exist was articulated in the Feder–Vardi program [5], and later algebraic work of Barto and Kozik clarified why finite structural boundaries are plausible in that setting [2]. The present paper should be read against that background. It does not argue against dichotomy theorems in general. It shows instead that, for exact relevance certification, quotient realizability and closure-law invariance leave too much room for a naive admissible classifier to survive. Any successful frontier theorem in this domain must therefore use stronger representation-sensitive structure than the first natural candidate provides.

Meta-Impossibility Traditions

The negative theorem in this paper belongs to a broader family of impossibility results. At the foundational level, Gödel’s incompleteness theorems show that sufficiently expressive formal systems cannot internalize all truths about their own proof theory [8]. In learning and optimization, no-free-lunch results show that universal guarantees disappear on overly broad problem domains [21, 22]. In social choice, Arrow’s theorem and the Gibbard–Satterthwaite theorems show that natural axiomatic requirements become jointly unsatisfiable on rich enough domains [1, 7, 15]. The present theorem has the same broad form: one fixes a principled invariance package and then proves that no exact classifier can satisfy it on the intended domain.

The closest analogy is nevertheless Rice-style impossibility. Rice’s theorem [14] shows that nontrivial extensional properties of sufficiently expressive semantic objects are undecidable. Refinements such as the Myhill–Shepherdson theorem and the Rice–Shapiro theorem show that this is not a single isolated fact but part of a broader pattern for structured semantic domains [12, 17]. The analogue in the present paper is that dominant-pair concentration, margin-boundedness, ghost-action concentration, and additive/statewise offset concentration are nontrivial slice predicates that

are invariant under the theorem-forced closure laws.

The Arrow parallel is complementary. The admissibility package plays the role of the axiom system: closure-law invariance formalizes semantic irrelevance, while polynomial-time checkability, structural extractability, and bounded-pattern definability exclude semantic encodings that would make the statement vacuous. Once those requirements are imposed, the orbit-gap witnesses show that exact correctness is impossible. The result is therefore best understood not as a pathology of one chosen classifier, but as a structural impossibility theorem for a natural class of representation-sensitive classifiers.

8 Conclusion

The paper now establishes both sides of the frontier question. On the positive side, the current formalized tractable landscape factors into primitive mechanisms, lifts, and degeneracies. On the negative side, quotient realizability is maximal, so quotient shape cannot classify tractability by itself, and an admissibility-layer collapse theorem rules out naive finite classifiers even after imposing theorem-forced closure laws together with computational and structural restrictions.

Technically, the decisive mechanism is closure-orbit disagreement via pair-targeted action-independent affine transport. The same template now applies to four obstruction families (dominant pair, margin masking, ghost action, and additive/statewise offset), yielding a formal admissible-collapse theorem.

The open problem is therefore sharper than before. The question is no longer whether simple finite lists or unconstrained quotient predicates can characterize the boundary; they cannot. The research program is redirected toward a narrower target: identify a stronger representation-sensitive principle, outside the present admissible surface but still mathematically natural, that could support a genuine optimizer-compatible dichotomy. In that sense, the meta-impossibility theorem is not only a negative result. It determines which kinds of frontier theorem are no longer worth pursuing.

9 Artifact Index and Supplementary Tables

This paper uses the same theorem-local proof-handle convention as the rest of the series: inline markers of the form `\LH{...}` and `\LHrng{...}` attach formal support directly to the corresponding paper claims.

The cited-handle closure comprises 11933 lines and 572 theorem-level declarations across 52 files; this is the code needed for the handles cited in the paper. The local development contains 9123 lines and 428 theorem-level declarations across 47 files, while the cumulative transitive dependency closure contains 80936 lines and 3519 theorem-level declarations across 310 files. Thus the local development contributes 11% of cumulative lines and 12% of cumulative theorem count. The supplementary handle ledger records source-file provenance for every cited handle: entries under `Frontier/*` are newly introduced here, whereas entries under `DecisionQuotient/*` are inherited from an earlier decision-quotient development.

The full handle ledger and claim-to-handle mapping are provided in the supplementary PDF and archived artifact. We omit those longer tables from the manuscript body.

References

- [1] Kenneth J. Arrow. A difficulty in the concept of social welfare. *Journal of Political Economy*, 58(4):328–346, 1950.

- [2] Libor Barto and Marcin Kozik. Constraint satisfaction problems solvable by local consistency methods. *Journal of the ACM*, 61(1):3:1–3:19, 2014.
- [3] Christian Bessiere, Clement Carbonnel, Emmanuel Hebrard, George Katsirelos, and Toby Walsh. Detecting and exploiting subproblem tractability. In *Proceedings of the Twenty-Third International Joint Conference on Artificial Intelligence (IJCAI)*, pages 468–474, 2013.
- [4] Andrei A. Bulatov. A dichotomy theorem for nonuniform csps. In *2017 IEEE 58th Annual Symposium on Foundations of Computer Science (FOCS)*, pages 319–330. IEEE, 2017.
- [5] Tomás Feder and Moshe Y. Vardi. The computational structure of monotone monadic snp and constraint satisfaction: A study through datalog and group theory. *SIAM Journal on Computing*, 28(1):57–104, 1998.
- [6] Robert Ganian, M. S. Ramanujan, and Stefan Szeider. Discovering archipelagos of tractability for constraint satisfaction and counting. In *Proceedings of the 27th Annual ACM-SIAM Symposium on Discrete Algorithms (SODA)*, pages 1670–1681. SIAM, 2016.
- [7] Allan Gibbard. Manipulation of voting schemes: A general result. *Econometrica*, 41(4):587–601, 1973.
- [8] Kurt G"odel. "Uber formal unentscheidbare s"atze der Principia Mathematica und verwandter systeme I. *Monatshefte f"ur Mathematik und Physik*, 38(1):173–198, 1931.
- [9] Isabelle Guyon and André Elisseeff. An introduction to variable and feature selection. *Journal of Machine Learning Research*, 3:1157–1182, 2003.
- [10] Ron Kohavi and George H. John. Wrappers for feature subset selection. *Artificial Intelligence*, 97(1-2):273–324, 1997.
- [11] Scott M. Lundberg and Su-In Lee. A unified approach to interpreting model predictions. In *Advances in Neural Information Processing Systems 30 (NeurIPS)*, pages 4765–4774, 2017.
- [12] John Myhill and John C. Shepherdson. Effective operations on partial recursive functions. *Zeitschrift f"ur mathematische Logik und Grundlagen der Mathematik*, 1(5):310–317, 1955.
- [13] Zdzisław Pawlak. Rough sets. *International Journal of Parallel Programming*, 11(5):341–356, 1982.
- [14] Henry Gordon Rice. Classes of recursively enumerable sets and their decision problems. *Transactions of the American Mathematical Society*, 74(2):358–366, 1953.
- [15] Mark Allen Satterthwaite. Strategy-proofness and Arrow’s conditions: Existence and correspondence theorems for voting procedures and social welfare functions. *Journal of Economic Theory*, 10(2):187–217, 1975.
- [16] Thomas J. Schaefer. The complexity of satisfiability problems. In *Proceedings of the Tenth Annual ACM Symposium on Theory of Computing*, pages 216–226. ACM, 1978.
- [17] Norman Shapiro. Degrees of computability. *Transactions of the American Mathematical Society*, 82(2):281–299, 1956.
- [18] Tristan Simas. The optimizer quotient and the certification trilemma. Zenodo, 2025. Archived manuscript and Lean artifact.

- [19] Roman Slowinski and Daniel Vanderpooten. Decision under uncertainty: A rough set approach. *European Journal of Operational Research*, 80(2):277–289, 1995.
- [20] Ryan Williams, Carla P. Gomes, and Bart Selman. Backdoors to typical case complexity. In *Proceedings of the Eighteenth International Joint Conference on Artificial Intelligence (IJCAI)*, pages 1173–1178. Morgan Kaufmann Publishers Inc., 2003.
- [21] David H. Wolpert. The lack of a priori distinctions between learning algorithms. *Neural Computation*, 8(7):1341–1390, 1996.
- [22] David H. Wolpert and William G. Macready. No free lunch theorems for optimization. *IEEE Transactions on Evolutionary Computation*, 1(1):67–82, 1997.
- [23] Dmitriy Zhuk. A proof of csp dichotomy conjecture. In *2017 IEEE 58th Annual Symposium on Foundations of Computer Science (FOCS)*, pages 331–342. IEEE, 2017.